Nikhil Prabhu

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Education

University of California, San Diego (UCSD) San Diego, CA
B.S. in Computer Science | Cumulative GPA: 3.67 *Expected Graduation: June 2028*

Skills

Languages: Python, C++, TypeScript, JavaScript

Robotics Stack: ROS 2, perception/planning/control pipelines, sensor integration, embedded deployment, real-time systems

Perception + Learning: PyTorch, OpenCV, YOLOX, MediaPipe, PPO/MAPPO, behavioral cloning, sim-to-real transfer

Simulation + Hardware: MuJoCo, Gymnasium, NVIDIA Jetson, Arduino embedded systems, Docker, field iteration on physical hardware

Experience

Lead Researcher — Foot Drop Ankle-Foot Orthosis (Georgia Tech mmWave Lab) Atlanta, GA
IEEE ICHCI 2024 Presenter & Published Author *May 2024 – Nov 2024*

- Deployed a real-time EMG sensing + motor actuation control system on embedded Arduino hardware through iterative field testing (R = 0.968 correlation, 23% gait precision gain).
- First-author paper published and presented at IEEE ICHCI 2024 on sensor placement and closed-loop control for assistive devices.

Lead Researcher — Brain-Computer Interface (Stanford Lee-Messer Lab) Stanford, CA
Stanford University School of Medicine *Oct 2023 – Dec 2024*

- Built and benchmarked ML models converting noisy sensor streams into reliable real-time control commands; piloted with 12 users on shipped assistive hardware.

Undergraduate Research Assistant — AI for Circuit Invention, UCSD San Diego, CA
Research Assistant (with Prof. Prasad Gudem & Samvid AI) *March 2026 – Present*

- Built a closed-loop generate-simulate-evaluate pipeline integrating LLM APIs with Cadence Spectre tools; exploring RL for design-space search over simulation feedback.

Robotics & Manufacturing Projects

Motion — Industrial Manipulation Data Platform
Stanford × DeepMind Hackathon *2026*

- Built a crowdsourced pipeline for industrial manipulation clips (pick & place, valve turns, box stacking) with live MediaPipe pose overlay and instant quality scoring on contributor devices.
- Designed a Gemma 4 + Kimi K2 VLM verification stack to score task match, hand visibility, and motion completeness; exported LeRobot / Open X-Embodiment bundles for physical-AI training.

SODIUM — Embodied Robot on Real Hardware
1st Overall, DiamondHacks 2026 *April 2026*

- Deployed a ROS 2 robot stack on Jetson-class embedded compute with on-device perception, wake-word detection, and real-time control integrated with a cloud operator dashboard (Svelte 5 / Bun).

Battle Angel — Sim-to-Real Policy Training
Google Gemini 3.5 Hackathon *2026*

- Trained vectorized PPO policies for a Unitree G1 humanoid across eight MuJoCo environments with automated evaluation, unseen-scene rollouts, and GIF replay — direct experience bridging simulation limits to real-world deployment.

DroneOS
1st Overall, Bow Capital × DS3 × SIC Hackathon (UCSD) *2026*

- Built perception (YOLOX + Norfair), SLAM, and visual odometry on edge hardware; trained MAPPO swarm policies and validated sim-to-real transfer from simulation to autonomous flight.

Publications

- **Real-Time EMG Control for Assistive Devices** — IEEE ICHCI 2024.
- **Co-Heritability of Antibiotic Resistance in TB Evolution** — Bioinformatics & Biomedical Technology, 2025.